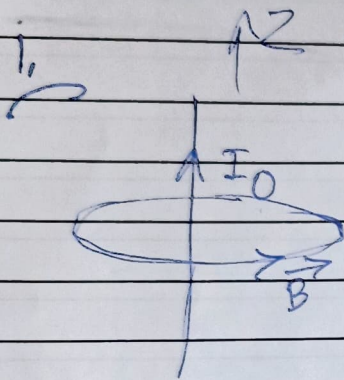




INTERNAL ASSESSMENT SHEET

Semester : ☐ Monsoon 20 ☐ Winter 20 ☐ Summer 20

Name : _____ Roll No. : _____
Course Code : _____ Course Title : _____
Date : _____ No. of Additional Sheet : _____
Student Sig : _____ Invigilator Sig : _____



Quiz 2

~~Ampere's Law~~

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I_0$$

$$\Rightarrow B 2\pi r = \mu_0 I_0 \Rightarrow B = \frac{\mu_0 I_0}{2\pi r}$$

[No part marking]

2. $\vec{B} = \nabla \times \vec{A}$

(4)
$$= \begin{vmatrix} \hat{r} & \hat{\theta} & \hat{\phi} \\ \frac{\partial}{\partial r} & \frac{1}{r} \frac{\partial}{\partial \theta} & \frac{1}{r \sin \theta} \frac{\partial}{\partial \phi} \\ 0 & 0 & 0 \end{vmatrix}$$

no part marks for $\vec{B}$

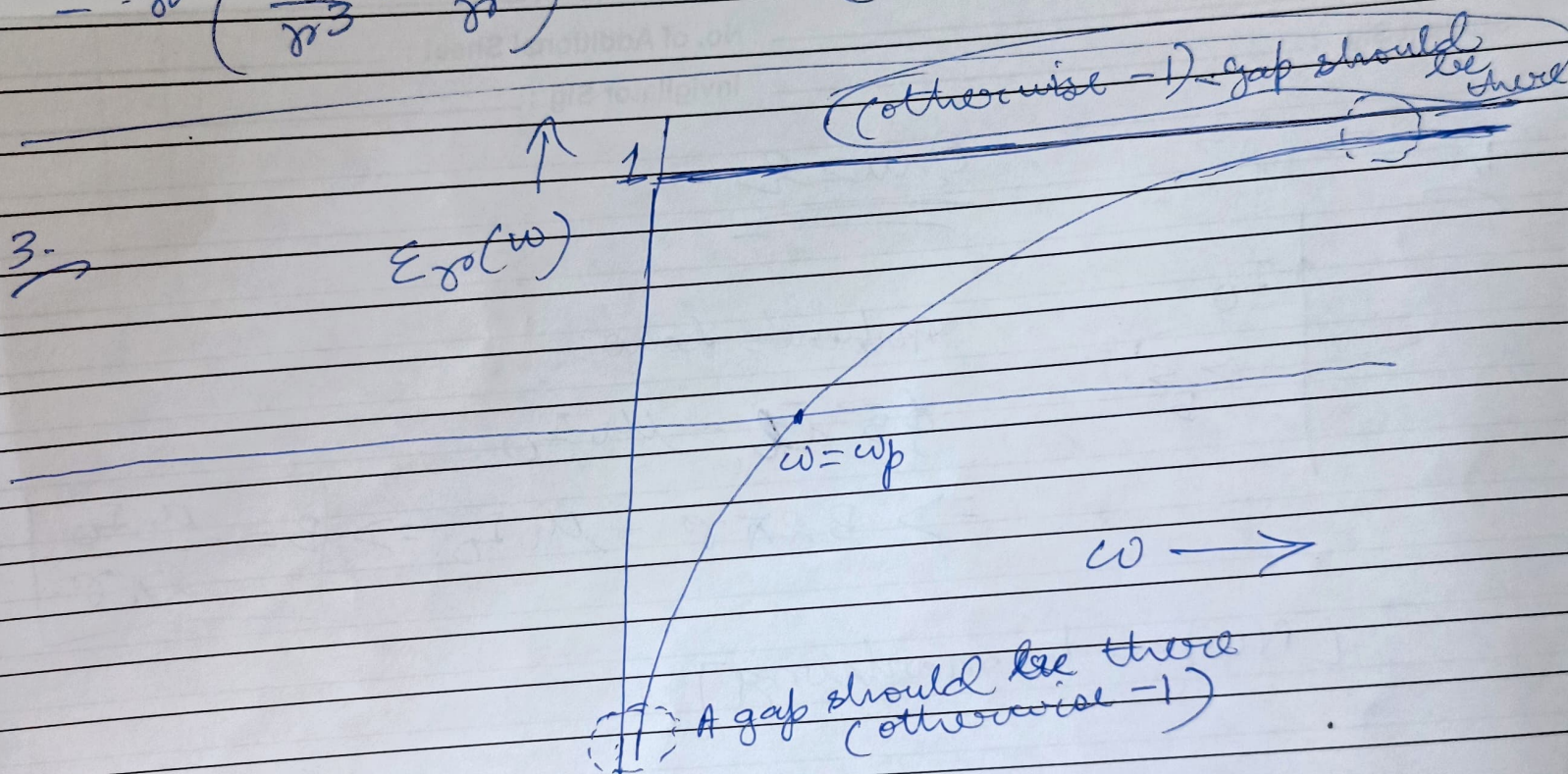
Some may not write the first row correctly. But that's okay as it is not required.

$$\vec{E} = -\nabla \phi - \frac{\partial \vec{A}}{\partial t} = \left\{ \frac{\partial}{\partial r} \left(\frac{1}{r^2} e^{i\omega t} \right) - \frac{\partial}{\partial t} \left\{ \frac{\partial}{\partial r} \frac{\partial}{\partial t} e^{i\omega t} \right\} \right\}$$

1 mark for writing this

$$= \frac{A}{r} \frac{2}{r^3} e^{i\omega t} - \frac{A}{r} i\omega e^{i\omega t}$$

$$= \frac{A}{r} \left(\frac{2}{r^3} - \frac{i\omega}{r} \right) e^{i\omega t} \quad (4)$$



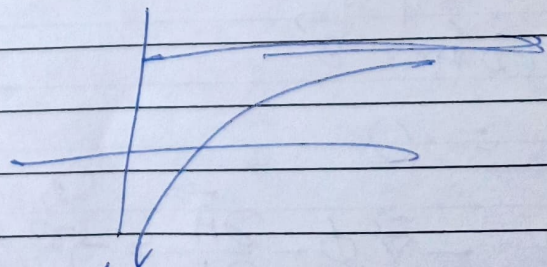
Axis label absent: -1

$\omega = \omega_p$ label u: -1

$E_r(\omega) = 1$ line is absent: -1

(Some may draw a symmetric graph for $\omega < 0$)
also: don't deduct marks

If someone draws like:



No marks.

No part marking.

4. (a) Polarization along x -axis (b) travelling along z -direction.

(c) $\omega = 5 \times 10^6$

$$c = 3 \times 10^8$$

$$k = \frac{\omega}{c} = \frac{5}{3} \times 10^{-2}$$

$$(d) \frac{E}{B} = c$$

$$\Rightarrow B = \frac{E}{c}$$

$$\therefore \vec{B} = \hat{y} \frac{4}{c} \exp\left\{i\left(5 \times 10^6 t - \frac{5}{3} \times 10^{-2} z\right)\right\}$$

(Must be a vector, otherwise no marks)
(Some may simplify by using calculator)
~~check and the values~~